COS 433: Cryptography

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Reminders:

Problem Set 0 due today, 11:59pm

Problem Set 1 release today, 11:59pm

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Secret-Key Encryption

Secret-Key Encryption: Syntax

$\text{ct} = \text{Enc}(k, m)$ →

$m' = \text{Dec}(k, \text{ct})$

k, m

k

Definition: A secret-key encryption scheme $(\textbf{Enc}, \textbf{Dec})$ consists of two algorithms:

- $\text{Enc}(k, m)$ takes a key $k \in \mathcal{K}$ and message $m \in \mathcal{M}$ and outputs a ciphertext $\text{ct} \in \mathcal{C}$
- $\text{Dec}(k, \text{ct})$ takes as input a key and ciphertext and outputs a message m'

A secret-key encryption scheme $(\textbf{Enc}, \textbf{Dec})$ is correct if for all $k \in \mathcal{K}, m \in \mathcal{M}$

$$\text{Dec}\big(k, \text{Enc}(k, m)\big) = m$$

Secret-Key Encryption: Security

k, m

$\mathrm{ct} = \mathrm{Enc}(k, m)$

$\longrightarrow$

k

Definition 1: a secret-key encryption scheme $(\mathrm{Enc}, \mathrm{Dec})$ is perfectly secret for uniform messages if for all functions $\mathrm{Eve}: \mathcal{C} \to \mathcal{M}$

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ m \leftarrow \mathcal{M}}}\left[\mathrm{Eve}\big(\mathrm{Enc}(k, m)\big) = m\right] = \frac{1}{|\mathcal{M}|}$$

Perfect Secrecy

Definition 1': a secret-key encryption scheme $(\mathbf{Enc}, \mathbf{Dec})$ is perfectly secret for uniform messages if for all messages $m^* \in \mathcal{M}$ and all ciphertexts $\mathbf{ct} \in \mathcal{C}$

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ m \leftarrow \mathcal{M}}} [m = m^* \mid \mathrm{Enc}(k, m) = \mathrm{ct}] = \frac{1}{|\mathcal{M}|}$$

This is equivalent to Definition 1. Why?

The best strategy for $\mathbf{Eve}$ given $\mathbf{ct}$ is to guess the message with the highest conditional probability of occurring.

Perfect Secrecy

Definition 1': a secret-key encryption scheme $(\mathbf{Enc}, \mathbf{Dec})$ is perfectly secret for uniform messages if for all messages $m^* \in \mathcal{M}$ and all ciphertexts $\mathbf{ct} \in \mathcal{C}$

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ m \leftarrow \mathcal{M}}} [m = m^* | \operatorname{Enc}(k, m) = \operatorname{ct}] = \frac{1}{|\mathcal{M}|}$$

But my messages aren't (uniformly) random!

Perfect Secrecy

Definition 2: a secret-key encryption scheme $(\mathbf{Enc}, \mathbf{Dec})$ is perfectly semantically secure if for all random variables M over messages, all messages $m^* \in \mathcal{M}$ and all $\mathrm{ct} \in \mathcal{C}$

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ M}}[M = m^* \mid \mathrm{Enc}(k, M) = \mathrm{ct}] = \Pr_{M}[M = m^*]$$

k, M

$$\xrightarrow{\mathrm{C} = \mathrm{Enc}(k, M)}$$

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Theorem: a secret-key encryption scheme is perfectly semantically secure if and only if it is perfectly secret for uniform messages!

Perfect Secrecy

Definition 3: a secret-key encryption scheme $(\mathrm{Enc}, \mathrm{Dec})$ is perfectly indistinguishable if for all message pairs (m_0, m_1)

$$\mathrm{Enc}(k, m_0) \equiv \mathrm{Enc}(k, m_1)$$

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Theorem: a secret-key encryption scheme is perfectly indistinguishable if and only if it is perfectly secret on uniform messages/perfectly semantically secure.

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Theorem: Definitions 1, 2, and 3 are equivalent

Proof: $2 \to 1 \to 3 \to 2$

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$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ M}}[M = m \mid \mathrm{Enc}(k, M) = \mathrm{ct}] = \Pr_{M}[M = m] = \frac{1}{|\mathcal{M}|}$$

Since $\mathbf{Eve}$ is only given $\mathbf{Enc}(k, M)$, she has no hope.

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Theorem: Definitions 1, 2, and 3 are equivalent

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Proof by contradiction. Suppose that $\exists (m_0, m_1)$ such that

$$\mathrm{Enc}(k, m_0) \not\equiv \mathrm{Enc}(k, m_1)$$

Then, $\exists\ \mathbf{ct}^*$ such that

$$\Pr_{k \leftarrow \mathcal{K}}[\mathrm{Enc}(k, m_0) = \mathrm{ct}^*] > \Pr_{k \leftarrow \mathcal{K}}[\mathrm{Enc}(k, m_1) = \mathrm{ct}^*]$$

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$\mathrm{Eve}(\mathrm{ct})$:

- If $\mathrm{ct} \neq \mathrm{ct}^*$, guess uniformly from $\mathcal{M}$.
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$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ M}}[M = m^* \wedge \mathrm{Enc}(k, M) = \mathrm{ct}] = \Pr_{M}[M = m^*] \cdot \Pr_{k,M}[\mathrm{Enc}(k, M) = \mathrm{ct}]$$

Perfect Secrecy

Theorem: Definitions 1, 2, and 3 are equivalent

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We know:

$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ M}}[M = m^* \wedge \mathrm{Enc}(k, M) = \mathrm{ct}] = \Pr_M[M = m^*] \cdot \Pr_k[\mathrm{Enc}(k, m^*) = \mathrm{ct}]$$

Claim: $$\Pr_k[\mathrm{Enc}(k, m^*) = \mathrm{ct}] = \Pr_{k,M}[\mathrm{Enc}(k, M) = \mathrm{ct}]$$

Proof: $$\forall m, \Pr_k[\mathrm{Enc}(k, m^*) = \mathrm{ct}] = \Pr_k[\mathrm{Enc}(k, m) = \mathrm{ct}]$$

Perfect secrecy seems like a good security notion!

The One-Time Pad

k, m

$\mathrm{ct} = \mathrm{Enc}(k, m)$

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$m' = \mathrm{Dec}(k, \mathrm{ct})$

Theorem: perfectly secret encryption schemes exist!

Simple proposal: let $\mathcal{K} = \mathcal{M} = \{0,1\}^n$ and define

$$\mathrm{Enc}(k, m) = k \oplus m$$

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Perfect indistinguishability: $\mathrm{Enc}(k, m) \equiv U_n$ (uniform n bit string)

The $\boxed{\text{One-Time}}$ Pad

$$\text{ct} = k \oplus m$$

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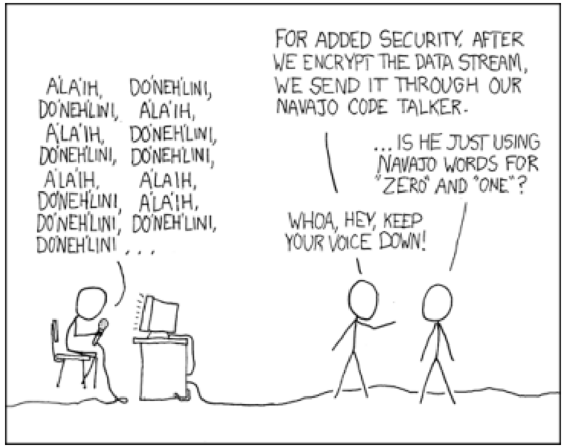
Why hasn't everyone been using this scheme all along?

$$\mathrm{Enc}(k, m_1) \oplus \mathrm{Enc}(k, m_2) = m_1 \oplus m_2$$



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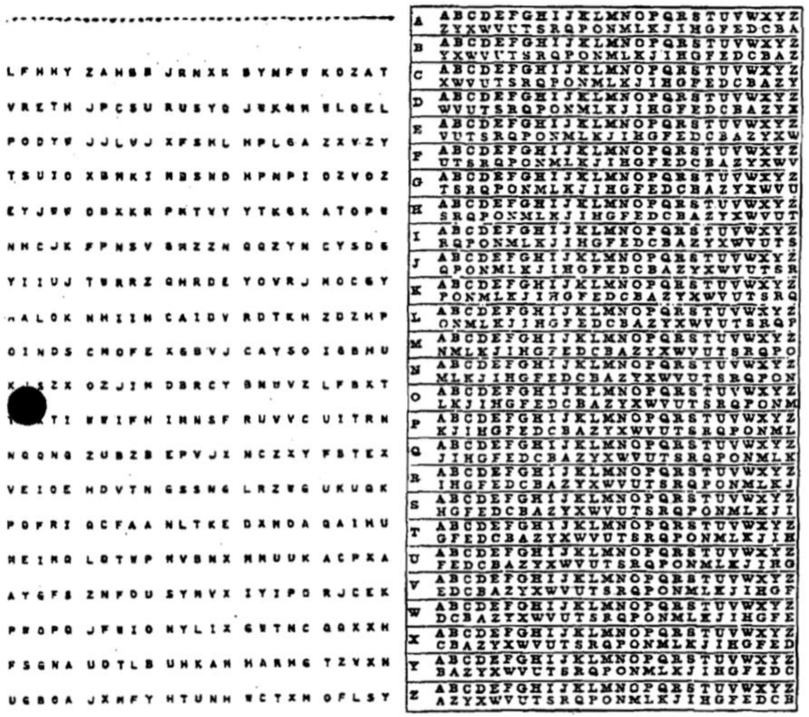

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Why hasn't everyone been using this scheme all along?

Corollary: If we want to (eventually) exchange n-bits worth of messages, have to share a key of length n

Can we do better?

- Encrypt *many* messages with perfect secrecy?
- Encrypt long messages with short keys?

No!

Theorem [Shannon]: Any perfectly secret encryption scheme satisfies

$$|\mathcal{K}| \geq |\mathcal{M}|$$

- Applies even if encryption is randomized.
- Extends immediately to "t-message perfect secrecy:"

$$|\mathcal{K}| \geq |\mathcal{M}|^t$$

Proof of Shannon’s Theorem

Intuition: if $|\mathcal{K}| < |\mathcal{M}|$, a uniformly random message “has too much information” to be hidden by a random key.

Proof: let M, M' denote independent, uniformly random messages from $\mathcal{M}$.

Claim: $$\big(M, \mathrm{Enc}(K, M)\big) \equiv \big(M, \mathrm{Enc}(K, M')\big)$$

This follows from perfect secrecy.

Proof of Shannon's Theorem

$$(M, \mathrm{Enc}(K, M)) \equiv (M, \mathrm{Enc}(K, M'))$$

Question: given a pair (m, ct) sampled in this way, what is the probability that <u>there exists a key</u> k such that $\mathrm{Dec}(k, \mathrm{ct}) = m$? ("Plausible pair")

$$\Pr_{K,M}[\exists k: \mathrm{Dec}(k, \mathrm{Enc}(K, M)) = M] \quad = 1 \qquad \text{(Correctness)}$$

Proof of Shannon's Theorem

$$\big(M, \mathrm{Enc}(K, M)\big) \equiv \big(M, \mathrm{Enc}(K, M')\big)$$

Question: given a pair (m, ct) sampled in this way, what is the probability that <u>there exists a key</u> k such that $\mathrm{Dec}(k, \mathrm{ct}) = m$? ("Plausible pair")

$$\Pr_{K,M,M'}[\exists k \colon \mathrm{Dec}(k, \mathrm{Enc}(K, M')) = M]$$

$$\leq \sum_k \Pr_{K,M,M'}[\mathrm{Dec}(k, \mathrm{Enc}(K, M')) = M] \quad = \frac{|\mathcal{K}|}{|\mathcal{M}|}$$

Proof of Shannon's Theorem

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Conclusion:

$$1 \leq \frac{|\mathcal{K}|}{|\mathcal{M}|}$$

What do we do now?

Perfect Secrecy

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- Extends immediately to "t-message perfect secrecy:"

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Takeaway: perfect secrecy is too rigid. How can we weaken the definition?

Statistical Secrecy

Definition: a secret-key encryption scheme $(\mathbf{Enc}, \mathbf{Dec})$ is ϵ-statistically secure if for all message pairs (m_0, m_1)

$$\mathrm{Enc}(k, m_0) \approx_\epsilon \mathrm{Enc}(k, m_1)$$

For all "distinguishers" $D: \mathcal{C} \to \{0,1\}$, letting $C_b = \mathrm{Enc}(k, m_b)$,

$$|\mathbf{E}[D(C_0)] - \mathbf{E}[D(C_1)]| \leq \epsilon$$

$$\mathrm{ct} = \mathrm{Enc}(k, m_b)$$

Statistical Secrecy

Theorem [Shannon]: Any statistically secret encryption scheme still requires large keys!

Proof: Problem Set 1

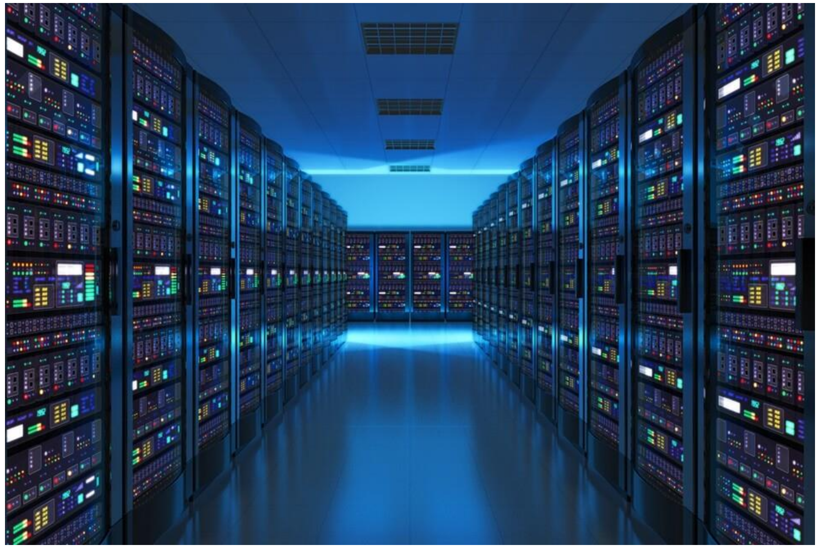
What do we do?

Idea: Assume the Adversary is Computationally Bounded

$$\xrightarrow{\mathrm{ct} = \mathrm{Enc}(k, m)}$$

k, m

k



(T, ϵ) Indistinguishability

Definition: Two random variables $X \approx_{T,\epsilon} Y$ are (T, ϵ)-computationally indistinguishable if for all "distinguishers" $D: \boldsymbol{\mathcal{X}} \to \{0,1\}$ computable in time T,

$$|\mathbf{E}[D(X)] - \mathbf{E}[D(Y)]| \le \epsilon$$

Definition: a one-time secret-key encryption scheme $(\mathbf{Enc}, \mathbf{Dec})$ is (T, ϵ)-computationally secure if for all message pairs (m_0, m_1)

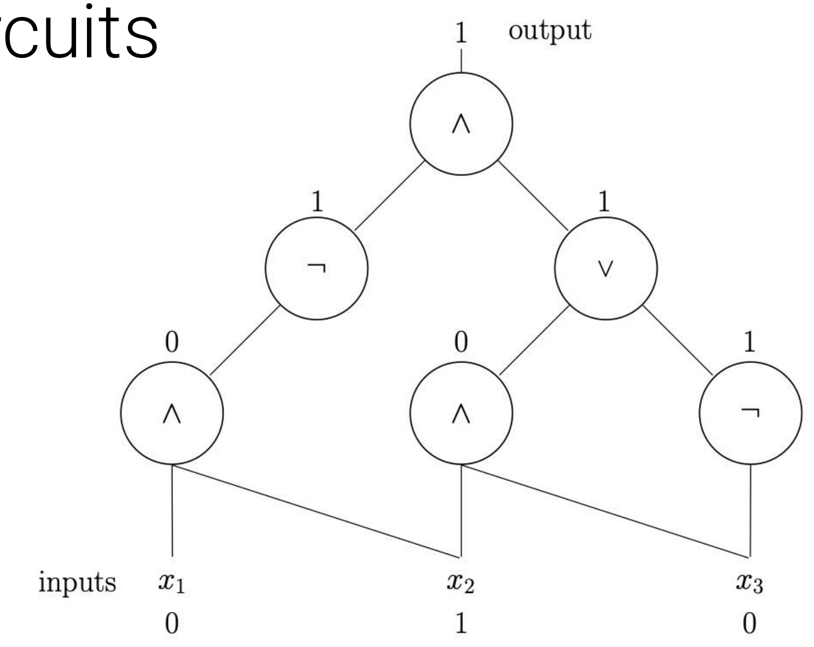
$$\mathrm{Enc}(k, m_0) \approx_{T,\epsilon} \mathrm{Enc}(k, m_1)$$

(T, ϵ) Indistinguishability

What does it mean for D to run in time T?

Adversary computational model: Boolean circuits

- T = number of gates in the circuit (size)
- Can simulate computation in your favorite programming language (with minimal slowdown).



(T, ϵ) Indistinguishability

General principle:

Assume that the adversary is more powerful than the honest parties.

k, m

$$\text{ct} = \text{Enc}(k, m) \longrightarrow$$

k

- Looking for (T, ϵ)-secure encryption schemes where the honest parties run in much less than T time.

- Real-world tradeoffs between efficiency and security.

(T, ϵ) Indistinguishability

Questions:

1. Is the modeling assumption valid? Depends on (T, ϵ).

2. Do (T, ϵ)-secure encryption schemes exist?

   - The modern world depends on it, but we don't know for sure!

3. How do we build such a scheme?

(T, ϵ) Indistinguishability

Questions:

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Time Estimation (don't try this at home)

Speed of a modern laptop: 2^{37} floating point operations per second.

Speed of a modern supercomputer: 2^{60} floating point operations per second.

1000 years $= 2^{35}$ seconds.

Age of the universe: $2^{58.5}$ seconds.

Can execute $\mathbf{2^{120}}$ floating point operations running a modern supercomputer for the current lifetime of the universe.

In the real world, usually set $T/\epsilon = 2^{128}, 2^{256}, 2^{512}$

Time Estimation (don't try this at home)

Concerns not addressed in this calculation:

- What about parallelization?
- Validity of the Boolean circuit computational model (Church-Turing thesis)
  - Now believed by many to be false (e.g., "quantum computers")
- "Future-proofing"

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Secret-Key Encryption

Secret-Key Encryption: Syntax

k, m

$$\xrightarrow{\mathrm{ct} = \mathrm{Enc}(k, m)}$$

k

$m' = \mathrm{Dec}(k, \mathrm{ct})$

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Secret-Key Encryption: Security

k, m

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Perfect Secrecy

Definition 1': a secret-key encryption scheme $(\mathbf{Enc}, \mathbf{Dec})$ is perfectly secret for uniform messages if for all messages $m^* \in \mathcal{M}$ and all ciphertexts $\mathbf{ct} \in \mathcal{C}$

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But my messages aren't (uniformly) random!

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Definition 2: a secret-key encryption scheme $(\mathbf{Enc}, \mathbf{Dec})$ is perfectly semantically secure if for all random variables M over messages, all messages $m^* \in \mathcal{M}$ and all $\mathrm{ct} \in \mathcal{C}$

$$\Pr_{\substack{k\leftarrow\mathcal{K}\\ M}}[M = m^* \mid \mathrm{Enc}(k, M) = \mathrm{ct}\,] = \Pr_{M}[M = m^*]$$

k, M

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Theorem: a secret-key encryption scheme is perfectly semantically secure if and only if it is perfectly secret for uniform messages!

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Proof by contradiction. Suppose that $\exists (m_0, m_1)$ such that

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$$\Pr_{\substack{k \leftarrow \mathcal{K} \\ M}}[M = m^* \wedge \mathrm{Enc}(k, M) = \mathrm{ct}] = \Pr_{M}[M = m^*] \cdot \Pr_{k,M}[\mathrm{Enc}(k, M) = \mathrm{ct}]$$

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We know:

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Claim: $$\Pr_k[\mathrm{Enc}(k, m^*) = \mathrm{ct}] = \Pr_{k,M}[\mathrm{Enc}(k, M) = \mathrm{ct}]$$

Proof: $$\forall m, \Pr_k[\mathrm{Enc}(k, m^*) = \mathrm{ct}] = \Pr_k[\mathrm{Enc}(k, m) = \mathrm{ct}]$$

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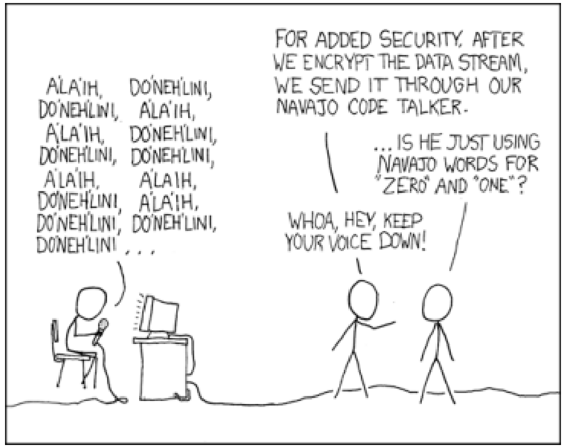

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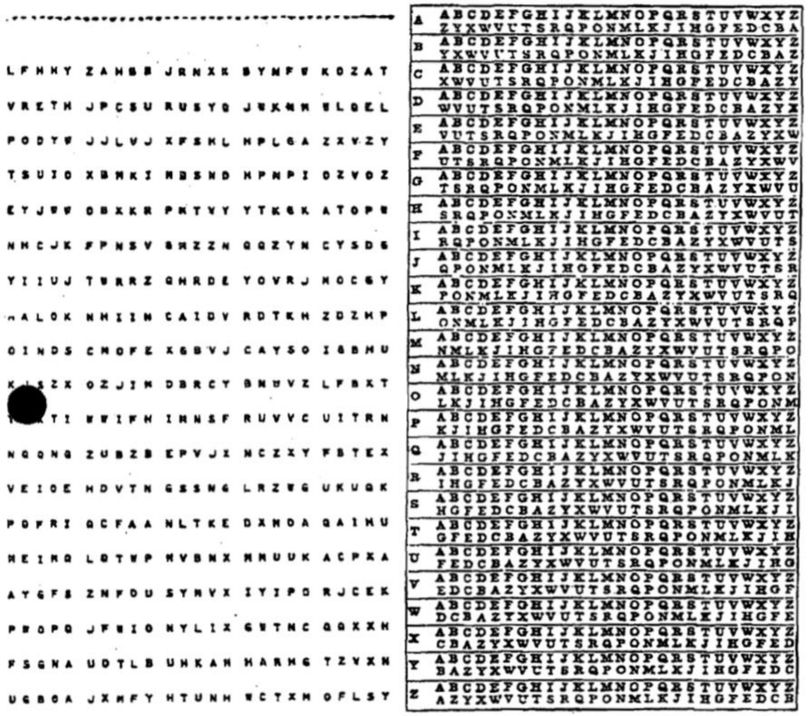

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Question: given a pair (m, ct) sampled in this way, what is the probability that <u>there exists a key</u> k such that $\mathrm{Dec}(k, \mathrm{ct}) = m$? ("Plausible pair")

$$\Pr_{K,M,M'}[\exists k\colon \mathrm{Dec}(k, \mathrm{Enc}(K, M')) = M]$$

$$\leq \sum_k \Pr_{K,M,M'}[\mathrm{Dec}(k, \mathrm{Enc}(K, M')) = M] \quad = \frac{|\mathcal{K}|}{|\mathcal{M}|}$$

Proof of Shannon's Theorem

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Question: given a pair (m, ct) sampled in this way, what is the probability that <u>there exists a key</u> k such that $\mathrm{Dec}(k, \mathrm{ct}) = m$? ("Plausible pair")

Conclusion:

$$1 \leq \frac{|\mathcal{K}|}{|\mathcal{M}|}$$

What do we do now?

Perfect Secrecy

Theorem [Shannon]: Any perfectly secret encryption scheme satisfies

$$|\mathcal{K}| \geq |\mathcal{M}|$$

- Applies even if encryption is randomized.
- Extends immediately to "t-message perfect secrecy:"

$$|\mathcal{K}| \geq |\mathcal{M}|^t$$

Takeaway: perfect secrecy is too rigid. How can we weaken the definition?

Statistical Secrecy

Definition: a secret-key encryption scheme $(\mathbf{Enc}, \mathbf{Dec})$ is ϵ-statistically secure if for all message pairs (m_0, m_1)

$$\mathrm{Enc}(k, m_0) \approx_\epsilon \mathrm{Enc}(k, m_1)$$

For all “distinguishers” $D: \mathcal{C} \to \{0,1\}$, letting $C_b = \mathrm{Enc}(k, m_b)$,

$$|\mathbf{E}[D(C_0)] - \mathbf{E}[D(C_1)]| \le \epsilon$$

$$\xrightarrow{\mathrm{ct} = \mathrm{Enc}(k, m_b)}$$

Statistical Secrecy

Theorem [Shannon]: Any statistically secret encryption scheme still requires large keys!

Proof: Problem Set 1

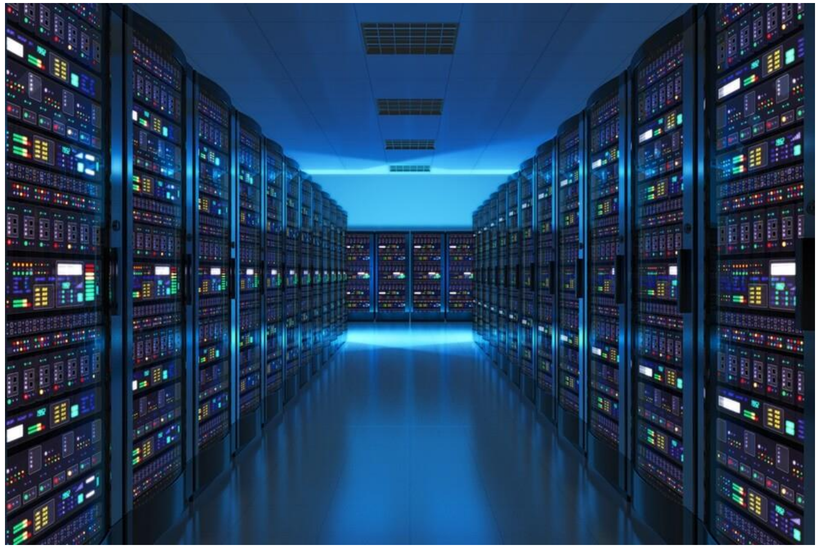
What do we do?

Idea: Assume the Adversary is Computationally Bounded

k, m

$$\mathrm{ct} = \mathrm{Enc}(k, m)$$

k



(T, ϵ) Indistinguishability

Definition: Two random variables $X \approx_{T,\epsilon} Y$ are (T, ϵ)-computationally indistinguishable if for all "distinguishers" $D: \boldsymbol{\mathcal{X}} \to \{0,1\}$ computable in time T,

$$|\mathbf{E}[D(X)] - \mathbf{E}[D(Y)]| \leq \epsilon$$

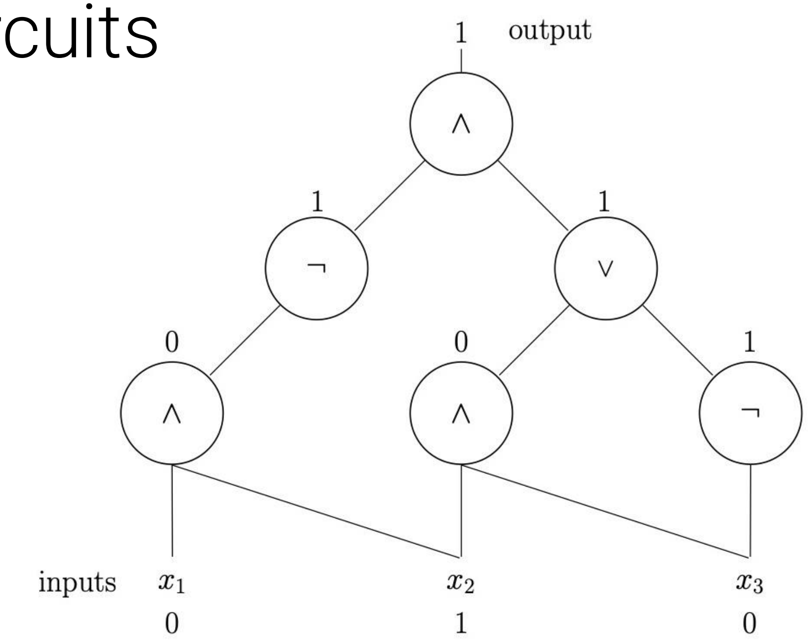
Definition: a one-time secret-key encryption scheme $(\mathrm{Enc}, \mathrm{Dec})$ is (T, ϵ)-computationally secure if for all message pairs (m_0, m_1)

$$\mathrm{Enc}(k, m_0) \approx_{T,\epsilon} \mathrm{Enc}(k, m_1)$$

(T, ϵ) Indistinguishability

What does it mean for D to run in time T?

Adversary computational model: Boolean circuits



- T = number of gates in the circuit (size)
- Can simulate computation in your favorite programming language (with minimal slowdown).

(T, ϵ) Indistinguishability

General principle:

Assume that the adversary is more powerful than the honest parties.

k, m

$\xrightarrow{\mathrm{ct} = \mathrm{Enc}(k, m)}$

k

- Looking for (T, ϵ)-secure encryption schemes where the honest parties run in much less than T time.

- Real-world tradeoffs between efficiency and security.

(T, ϵ) Indistinguishability

Questions:

1. Is the modeling assumption valid? Depends on (T, ϵ).

2. Do (T, ϵ)-secure encryption schemes exist?

   - The modern world depends on it, but we don't know for sure!

3. How do we build such a scheme?

(T, ϵ) Indistinguishability

Questions:

1. Is the modeling assumption valid? Depends on (T, ϵ).

Time Estimation (don't try this at home)

Speed of a modern laptop: 2^{37} floating point operations per second.

Speed of a modern supercomputer: 2^{60} floating point operations per second.

1000 years $= 2^{35}$ seconds.

Age of the universe: $2^{58.5}$ seconds.

Can execute $\mathbf{2^{120}}$ floating point operations running a modern supercomputer for the current lifetime of the universe.

In the real world, usually set $T/\epsilon = 2^{128}, 2^{256}, 2^{512}$

Time Estimation (don't try this at home)

Concerns not addressed in this calculation:

- What about parallelization?
- Validity of the Boolean circuit computational model (Church-Turing thesis)
  - Now believed by many to be false (e.g., "quantum computers")
- "Future-proofing"

In the real world, usually set $T/\epsilon = 2^{128}, 2^{256}, 2^{512}$